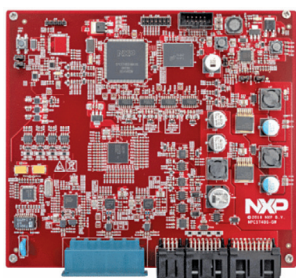


Secure Ethernet gateway design

The Ethernet gateway reference design from NXP Semiconductors is a development platform for creating Ethernet gateway electronic control units (ECUs) that operate securely and reliably. It is designed to accelerate development and supports multiple automotive communication protocols, enabling integration into vehicle networks. The design includes eight CAN FD (controller-are network flexible data-rate)



channels, expandable to twelve via SPI (serial peripheral interface), with data rates up to 5Mbps. For secure over-the-air (OTA) management, the board includes 4GB onboard eMMC memory, allowing remote firmware updates so vehicles can receive new features and security patches without in-person servicing, a requirement for modern automotive ECUs.

It also includes an embedded hardware security module (HSM) that supports secure hardware extension (SHE) and EVITA security standards to protect vehicle data and communications.

Applications. Key applications include industrial automation, smart grid systems, IoT networks, and automotive communications.

Link. <https://www.electronics-foru.com/electronics-projects/secure-ethernet-gateway-reference-design>

OEM Brand. NXP Semiconductors

Energy-efficient Ethernet designs

Broadcom Inc.'s reference designs provide a starting point for building 24-port Gigabit Ethernet copper

switches for both unmanaged and managed systems. You can develop Layer 2 switching products with standard features such as spanning tree, IEEE 802.1p/Q traffic prioritisation, and IPv4/IPv6 support, while testing switching behaviour, validating packet handling, and optimising throughput across 24 ports of 10/100/1000Mb/s Ethernet.

For unmanaged switches, the designs enable quick development of cost-sensitive products with basic forwarding performance and optimised power consumption for plug-and-play environments. For



managed switches, you can configure 8-, 16-, or 24-port systems, integrate management features, and test user interfaces, remote management, VLANs, traffic control, firmware updates, diagnostics, and network monitoring.

Applications. The reference designs are suitable for home networks, small business servers, and embedded networking systems

Link. <https://www.electronics-foru.com/electronics-projects/energy-efficient-ethernet-reference-designs>

OEM Brand. Broadcom

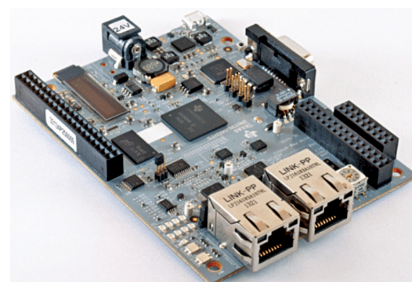
Design for industrial Ethernet

The reference design from Texas Instruments (TI) provides a pre-validated, integrated platform combining industrial Ethernet PHYs, a PROFINET IRT switch, a PROFINET IRT stack, and application examples, helping you reduce development time, lower risk, and accelerate time-to-market for industrial Ethernet devices. The design includes real-time features such as a 250µs cycle time and one-step time synchronisation (PTCP), eight consumer and provider protocol machines for efficient data exchange, and media redundancy

protocol (MRP) for network continuity.

Standard communication stacks, such as the Moxex PROFINET stack and the Intermiche SNMP stack, are integrated, allowing you to focus on application-specific development rather than on low-level protocols.

Applications. This reference design can be used for industrial automation, factory equipment, robotics, process control, PROFINET gateways, and Industry 4.0 smart manufacturing systems.



Link. <https://www.electronics-foru.com/electronics-projects/reference-design-for-industrial-ethernet>

OEM Brand. Texas Instruments (TI)

The above-mentioned Ethernet reference designs provide ready-to-use frameworks for building a wide range of networked systems, from high-density switches and energy-efficient solutions to secure gateways, industrial Ethernet devices, and robust front-end interfaces.

They help reduce development time, simplify integration, and ensure compliance with standards while supporting applications across enterprise networks, data centres, automotive systems, IoT, and industrial automation. These platforms enable focus on differentiation at the system level while reducing the effort required for foundational network design, validation, and standards compliance. **EFY**

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